

FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2022
FULL TEST – VII
PAPER –2
TEST DATE: 07-08-2022

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

Section – A

1. A, D

Sol. $V_{PQ} = \frac{-d\phi}{dt} = -e^t \left(\frac{\sqrt{3}}{2} a^2 - \frac{\pi a^2}{6} \right)$

$$V_{PQ} = V_{QR} = \frac{V_{PR}}{2}$$

$$V_{OP} = 0$$

2. A, C

Sol. $\Delta Q = ms\Delta T + \Delta U_{\text{electrostatic}}$

$$\Delta U_{\text{electrostatic}} = -\frac{kQ^2}{2R^2} \Delta R$$

3. A, D

Sol. $B_{\text{coil}} = \frac{\mu_0 i}{2R}$

$$dB_{\text{coil}} = -\frac{\mu_0 i}{2R^2} dR, \text{ here } dR = d$$

4. A, D

Sol. $F_{\text{net}} = m\omega^2 r - Bq\omega r < 0$

$$\omega < \frac{Bq}{m}$$

5. B, C

Sol. $\sin i = n \sin 30^\circ$
 $(n + \Delta n) \sin 30^\circ = \sin e$
Total deviation
 $\delta = i + e - 60^\circ$

6. A, C

 Sol. Number of carbon atoms in 4g carbon of living tree = $\frac{4}{12} \times 6 \times 10^{23} \times 8 \times 10^{-14} = 16 \times 10^9$

$$\text{Number of carbon atoms at present} = \frac{16 \times 10^9}{2^{t/T}} = \frac{RT}{\ln 2} = \frac{1}{3} \times \frac{2.1 \times 10^9}{0.7}$$

$$\frac{t}{T} = 4$$

7. C

8. D

Sol. (for Q. 7-8)

In first case

$$\vec{F} = -b\vec{v}$$

$$\Rightarrow \Delta \vec{p} = -b\vec{S}_1$$

In second case

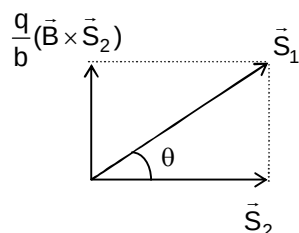
$$\vec{F} = -b\vec{v} + q(\vec{v} \times \vec{B})$$

$$\Delta \vec{p} = -b\vec{S}_2 + q(\vec{S}_2 \times \vec{B})$$

$$\text{Solving } b(\vec{S}_2 - \vec{S}_1) = q(\vec{S}_2 \times \vec{B})$$

$$\theta = \sin^{-1} \left(\frac{BqS_2}{bS_1} \right)$$

Heat does not depend on magnetic field.



9. A

10. C

Sol. (for Q. 9-10)

Force on dipole at x distance from center of ring

$$F = M \frac{dB}{dx}$$

$$F = -\frac{3}{2} \frac{M\mu_0 i R^2 x}{(R^2 + x^2)^{5/2}}$$

 For $x \ll R$

$$F = -\left(\frac{3M\mu_0 i}{2R^3} \right) x$$

$$T = 2\pi \sqrt{\frac{2mR^3}{3M\mu_0 i}}$$

$$\text{Potential energy} = -MB = -M \frac{\mu_0 i}{2R}$$

 For escape $K + U = 0$

$$\Rightarrow V = \sqrt{\frac{M\mu_0 i}{Rm}}$$

Section – B

11. 6

Sol. $i = 3r$

$$n \sin r = \sin i = \sin 3r$$

$$n = 3 - 4 \sin^2 r$$

$$dn = -4 \sin 2r \, dr$$

$$|dn| \text{ is maximum for } r = 45^\circ \Rightarrow i = \frac{3\pi}{4} \text{ radian}$$

12. 58

Sol. Speed of pulse

$$\sqrt{10} = \sqrt{\frac{T}{\mu}}$$

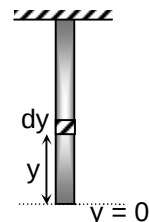
$$T = 10\mu$$

$$dT = 10d\mu$$

$$\frac{d\mu}{\mu} = dy$$

$$\mu = \mu_0 e^y = e^y$$

$$y_{\text{car}} = \frac{\int_0^1 e^y \cdot y \, dy}{\int_0^1 e^y \, dy} = 0.58$$



13. 1

Sol. In new system

$$1M = 15 \text{ kg}$$

$$\text{Then } G = \frac{20}{3} \times 10^{-11} \frac{M^3}{\text{kgs}^2} = 1 \frac{L^3}{\text{Ms}^2}$$

$$\Rightarrow 15 \times \frac{2}{3} \times 10^{-10} M^3 = L^3$$

$$\Rightarrow L = 10^{-3} M$$

Section – C

14. 00000.62

15. 00000.00

Sol. (for Q. 14-15)

If block are pulled down by x spring has $3x$ extension so net force on system – $3T = 9kx$

$$\text{So, time period} = 2\pi \sqrt{\frac{3m}{9k}} = 2\pi \sqrt{\frac{m}{3k}}$$

There is no tendency of slipping between them, hence friction is always zero.

16. 00000.41

17. 00001.82

Sol. (for Q.16-17)
Capacitance of small strip

$$dC = \frac{A\epsilon_0}{dx} \left(1 + \frac{x}{d} \right)$$

For some capacitance

$$\int_0^x \frac{1}{dC} = \int_x^d \frac{1}{dC}$$

$$[\log(d+x)]_0^x = [\log(d+x)]_x^d$$

$$\Rightarrow \frac{d+x}{d} = \frac{2d}{d+x}$$

$$\Rightarrow x^2 + 2dx - d^2 = 0$$

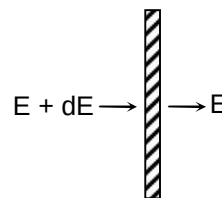
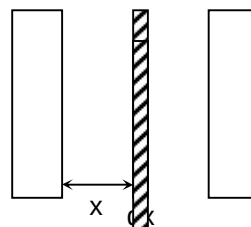
$$X = 0.41 d$$

Gauss law

$$dE \cdot A = \frac{\rho A dx}{\epsilon_0}$$

$$\rho = \epsilon_0 \frac{dE}{dx}$$

$$\rho \propto \frac{1}{(x+d)^2}$$



18. 00000.52

19. 00000.78

Sol. (for Q. 18-19)

$$R = \frac{2u^2 \sin \theta \cos \theta}{g}$$

Impulse of normal on brick at landing = $\mu u \sin \theta$

Impulse due to friction = $\mu \mu u \sin \theta = \mu u \cos \theta - mv_x$

$$v_x = u(\cos \theta - \mu \sin \theta)$$

$$\text{Horizontal distance covered} = R + \frac{v_x^2}{2\mu g}$$

$$= \frac{2u^2 \sin \theta \cos \theta}{g} + \frac{u^2 (\cos \theta - \mu \sin \theta)^2}{2\mu g}$$

$$d = \frac{u^2}{2\mu g} (\cos \theta + \mu \sin \theta)^2$$

For maximum distance $\mu = \tan \theta$

But this is valid only for $v_x > 0$

$$\Rightarrow \cos \theta - \mu \sin \theta > 0$$

$$\tan \theta < \frac{1}{\mu}$$

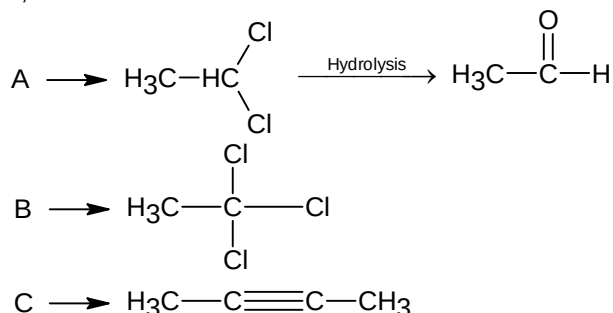
Chemistry

PART – II

Section – A

20. A, C

Sol.



21. A, C

Sol. Decarboxylation reaction takes place via carbanion intermediate.

22. C, D

Sol. At point B,

$$PV = nRT$$

$$V = \frac{nRT}{P} = \frac{1 \times 0.082 \times 600}{3}$$

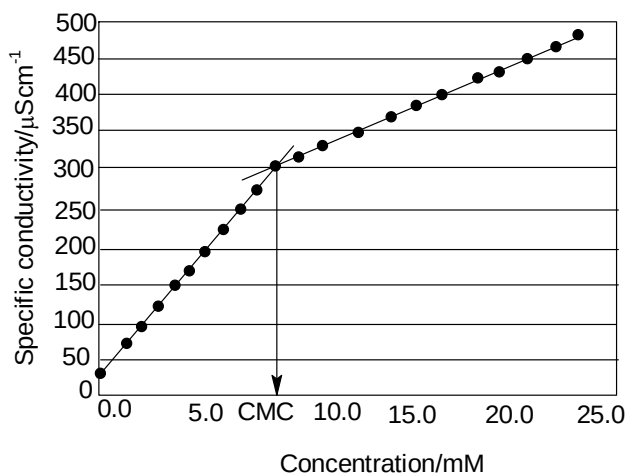
$$= 16.42 \text{ lit.}$$

$$\text{Also as pressure is same at A and C so } \frac{V_C}{V_A} = \frac{T_C}{T_A} = \frac{3}{1}$$

$$\therefore V_C = 3 \times V_A$$

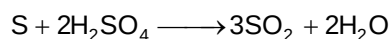
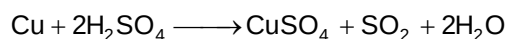
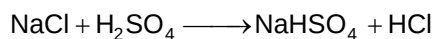
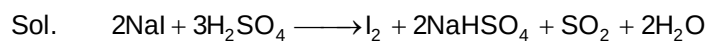
23. B, C, D

Sol. When KI is added to excess of AgNO_3 solution, positively charge sol is obtained due to adsorption of Ag^+ ions from dispersion medium. Increasing the length of the hydrocarbon chain increases the Kraft temperature.

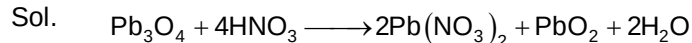


Specific conductivity versus surfactant solution concentration showing linear fits below and above the inflection point and the extrapolated intercept, which is the CMC. Chemisorption is favourable at high temperature.

24. A, C, D

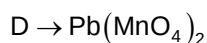
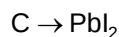


25. B, C

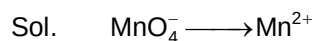


(A)

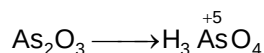
(B)



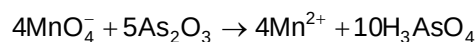
26. A



$nf = 5$



$nf = 4$



27. C



$200 \times 0.01 \times 5 = 10 \times N$

$N = 1$

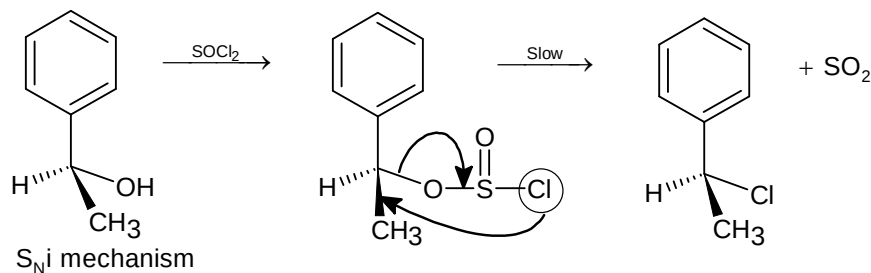
$\text{Volume strength} = 5.6$

$\text{Volume of O}_2 = 5.6 \times 20 \text{ ml}$

$= 112 \text{ ml}$

28. B

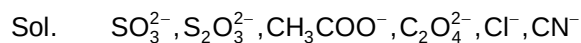
Sol.



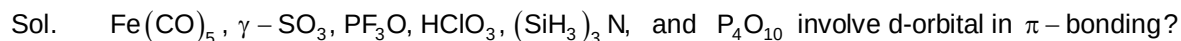
29. B

Section – B

30. 6

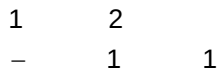
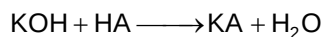


31. 6



32. 5

Sol. After electrolysis $[\text{KOH}] = \frac{96.5 \times 1}{96500 \times 0.1}$
 $= 10^{-2}$



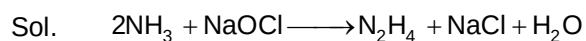
It will be a buffer solution

$$\text{pH} = \text{pK}_a + \log \frac{1}{1}$$

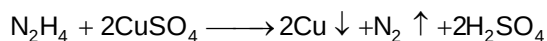
$$= \text{pK}_a = 5$$

Section – C

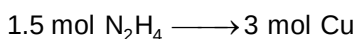
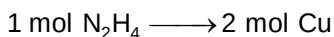
33. 00190.50



X



Y Z



$$= 3 \times 63.5$$

$$= 190.5 \text{ g}$$

34. 00001.33

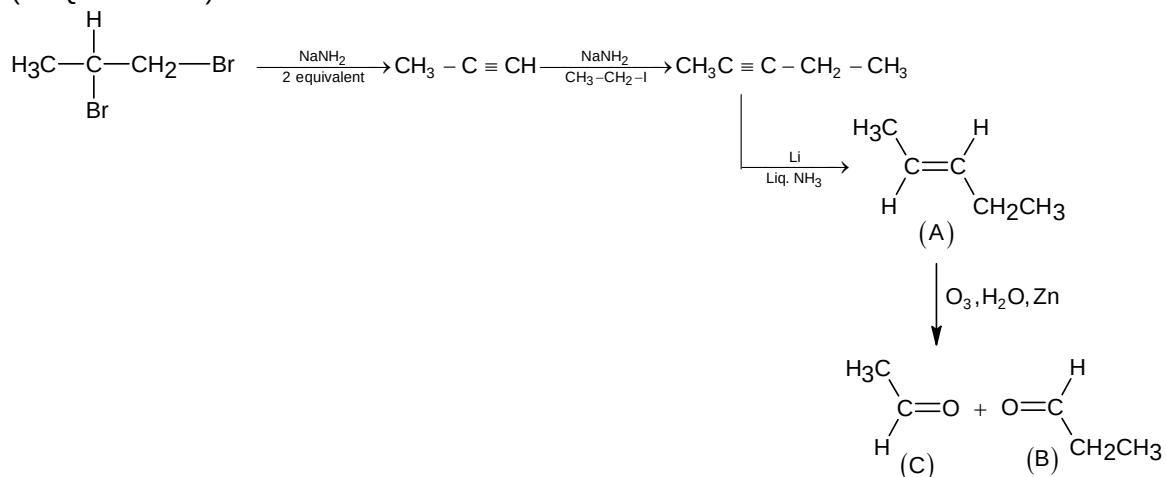
Sol. X is N_2H_4 , where oxidation state of N is – 2(a)

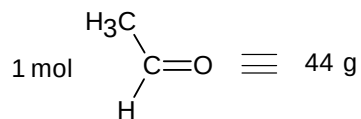
$$\therefore \left| \frac{2a}{3} \right| = \frac{2 \times 2}{3} = 1.33$$

35. 00058.50

36. 00035.20

Sol. (for Q.35 and 36)

From 1 mol 1,2-dibromo propane to pent-2-yne 3 mol NaNH_2 is needed.



$$\therefore 0.8 \text{ mol} = (44 \times 0.8) \text{ g} \\ = 35.20 \text{ g}$$

37. 00002.25

38. 00000.50

Sol. $T_A = \frac{1}{2}mv^2 \quad \lambda = \frac{h}{mv}$

$$= \frac{1}{2} \frac{m^2 v^2}{m} \\ = \frac{1}{2} \frac{h^2}{\lambda_B^2 m} \quad \dots (1)$$

$$T_B = \frac{1}{2} \frac{h^2}{\lambda_A^2 m} \quad \dots (2)$$

$$\therefore \frac{T_A}{T_B} = \frac{\lambda_B^2}{\lambda_A^2} = 4$$

Again

$$T_B = T_A - 1.50 \text{ eV}$$

$$T_B = 4T_B - 1.50 \text{ eV}$$

$$T_B = 0.50 \text{ eV}$$

$$T_A = 2.00 \text{ eV}$$

Now,

Work function of A

$$= 4.25 - T_A$$

$$= 2.25 \text{ eV}$$

Work function of B

$$= 4.20 - T_B$$

$$= 4.20 - 0.50$$

$$= 3.70$$

Mathematics**PART – III****Section – A**

39. A, D

Sol. $I = \int_{\pi/6}^{\pi/3} \left(e^{2\sin x} + e^{\frac{3}{2}\operatorname{cosec} x} \right) \cot x \, dx = \int_{1/2}^{\sqrt{3}/2} \left(\frac{e^{2x} + e^{\frac{3}{2x}}}{x} \right) dx$. Put $x = \frac{3}{4t}$

$$I = \int_{\sqrt{3}/2}^{3/2} \left(\frac{e^{2t} + e^{\frac{3}{2t}}}{t} \right) dt \Rightarrow 2I = \int_{1/2}^{3/2} \left(\frac{e^{2x} + e^{\frac{3}{2x}}}{x} \right) dx$$

$$I = \int_{1/2}^{\sqrt{3}/2} \frac{e^{2x} + e^{\frac{3}{2x}}}{x} dx > \int_{1/2}^{\sqrt{3}/2} \frac{2e^{2x}}{x} dx$$

$$\text{Also, } \int_{1/2}^{\sqrt{3}/2} \frac{2e^{2x}}{x} dx > \frac{4}{\sqrt{3}} \int_{1/2}^{\sqrt{3}/2} e^{2x} dx = \frac{2}{\sqrt{3}} (e^{\sqrt{3}} - e)$$

40. A, B, C, D

Sol. $3X - 4YY^T = I$

$$\therefore 3X^T - 4YY^T = I \Rightarrow X = X^T$$

$$\text{Also, } Y^{-1} = 2X^T = 2X$$

$$(Y^{-1})^T = 2X^T = Y^{-1}$$

$$\Rightarrow (Y^T)^{-1} = Y^{-1} \Rightarrow Y = Y^T$$

$$Y^{-1}Y = 2XY = I \text{ and } YY^{-1} = 2YX = I$$

$$3XY - 4Y^3 = Y \Rightarrow 4Y^3 + Y = \frac{3}{2}I$$

$$|2Y^T - Y + 4Y^3 - 5YX| = |-I| = -1$$

$$T_r((2X)^{-1} + I - 2XY + 4Y^3) = T_r(Y + 4Y^3) = \frac{9}{2}$$

41. B, C

Sol. (A) $P(E) = \frac{{}^{15}C_3}{{}^{16}C_4} = \frac{1}{4}$

(B) $P(E) = \frac{{}^{14}C_2}{{}^{15}C_3} = \frac{1}{5}$

(C) $P(E) = \frac{{}^{14}C_2}{{}^{15}C_3} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{20}$

(D) $P(E) = \frac{{}^{14}C_3}{{}^{15}C_4} = \frac{4}{15}$

42. B, D

Sol. $\sec^{-1} x = (4n \pm 1) \frac{\pi}{10} \Rightarrow x = \sec \frac{\pi}{10}, \sec \frac{3\pi}{10}, \sec \frac{7\pi}{10}, \sec \frac{9\pi}{10}$

43. A, B, C

$$\text{Sol. } \frac{2r+3}{\frac{3}{2}} = \frac{\lambda r+3}{3} = \frac{\lambda r}{-\frac{2}{2}} \Rightarrow \lambda = 1, r = -1$$

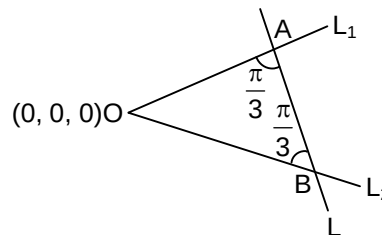
$$\therefore A \equiv (1, 2, -1) \because OA = OB$$

$$\Rightarrow 6r^2 + 18r + 12 = 0 \Rightarrow r = -1, -2$$

$$\therefore B \equiv (-1, 1, -2)$$

 $\therefore$ Equation of required plane containing

$$L_1 \text{ and } L_2 \text{ is } \begin{vmatrix} x & y & z \\ 1 & 2 & -1 \\ -1 & 1 & -2 \end{vmatrix} = 0$$



44. A, C

$$\text{Sol. } {}^{4n+2}C_{4r} = \frac{4n+2}{4r} {}^{4n+1}C_{4r-1} \Rightarrow \sum_{r=1}^n {}^{4n+1}C_{4r-1} = \frac{2}{2n+1} \sum_{r=1}^n {}^{4n+2}C_{4r}$$

$$\therefore (1+i)^{4n+2} = \sum_{k=0}^{4n+2} {}^{4n+2}C_k i^k$$

$$\therefore {}^{4n+2}C_0 - {}^{4n+2}C_2 + {}^{4n+2}C_4 - {}^{4n+2}C_6 + {}^{4n+2}C_8 - \dots = 0$$

$$\text{Also, } {}^{4n+2}C_0 + {}^{4n+2}C_2 + {}^{4n+2}C_4 + {}^{4n+2}C_6 + {}^{4n+2}C_8 + \dots = 2^{4n+1}$$

$$\Rightarrow \sum_{r=1}^n {}^{4n+2}C_{4r} = 2^{4n} - 1$$

$$\therefore f(n) = (2^{4n} - 1) \frac{2}{2n+1}$$

45. B

46. C

Sol. (for Q. 45-46)

$$xf(x) = 4 \int_0^x f(t) dt \Rightarrow f(x) = 2x^3$$

$$\text{Equation of tangent to the curve } y = 2x^3 \text{ is } y = 6t^2x - 4t^3$$

$$\Rightarrow m = 6t^2 \text{ and } -m = 4t^3 \therefore t = -\frac{3}{2} (t \neq 0) \therefore m = \frac{27}{2}$$

$$\int_0^1 (f(x) - g(x))g(x) dx = \frac{1}{7} \Rightarrow g(x) = x^3 \therefore \text{Required area} = \int_0^2 x^3 dx + \int_2^\infty 2^{5-x} dx = 4 + \frac{8}{\ln 2}$$

47. B

48. C

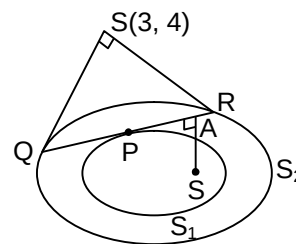
Sol. (for Q. 47-48)

 $\therefore$ Foot of perpendicular lies on auxiliary circle

$$\therefore a^2 = 9$$

$$\text{Equation of tangent at P is } y = -\frac{x}{2} + \frac{5}{2} \dots (1)$$

$$\therefore P \equiv \left(-\frac{a^2 m}{c}, \frac{b^2}{c} \right) \text{ i.e. } \left(\frac{9}{5}, \frac{8}{5} \right)$$



Equation of chord of contact of $S_2 = 0$ w.r.t S is

$$\frac{3x}{15} + \frac{4y}{b^2} = 1 \quad \dots (2)$$

$\Rightarrow b^2 = 10 \therefore S(3, 4)$ lies on director circle of $S_2 = 0$

$\therefore$ Also mid-point of QR is $\left(\frac{15}{11}, \frac{20}{11}\right) \therefore T$ is $\left(-\frac{3}{11}, -\frac{4}{11}\right)$

Section - B

49. 5

Sol. $2\sin\left(25x + \frac{\pi}{4}\right) = \sqrt{1 + 8\sin 50x \cos^2 50x} = \sqrt{1 + 4\sin 100x \cos 50x}$

$$2\sin\left(25x + \frac{\pi}{4}\right) = \sqrt{1 + 2(\sin 150x + \sin 50x)}$$

$$\Rightarrow 4\sin^2\left(\frac{\pi}{4} + 25x\right) = (1 + 2(\sin 150x + \sin 50x))$$

$$\Rightarrow \sin 150x = \frac{1}{2}; x = \left(n\pi + (-1)^n \frac{\pi}{6}\right) \frac{1}{150}$$

$$0 < n + (-1)^n \frac{1}{6} < 6 \text{ and } 0 < 6n + (-1)^n + 9 < 36$$

$$\Rightarrow n \in [0, 4]$$

$$\text{Also } 0 < 25x + \frac{\pi}{4} < \pi \Rightarrow 0 < \frac{\pi}{36}(6n + (-1)^n + 9) < \pi \Rightarrow 0 < 6n + (-1)^n + 9 < 36$$

$$0 \leq n \leq 4$$

50. 226

Sol. $\frac{15z-1}{z} = (-1)_{100} = e^{i(2r+1)\frac{\pi}{100}} \Rightarrow \frac{1}{z} = 15 - e^{i(2r+1)\frac{\pi}{100}}$

$$\frac{1}{z} = 15 - \cos\theta - i\sin\theta; \theta = (2r+1)\frac{\pi}{100}$$

$$\therefore \frac{1}{z} = 15 - \cos\theta + i\sin\theta \Rightarrow \left[\sum_{r=1}^{100} \frac{1}{|z_r|^2} \right] = \sum (226 - 30\cos\theta) = 22600$$

51. 50

Sol. $I_{n+1} - I_n = \int_0^\pi \frac{\sin\left(n + \frac{1}{2}\right)x}{\sin \frac{x}{2}} dx \quad \dots (1)$

$$\Rightarrow I_n - I_{n-1} = \int_0^\pi \frac{\sin\left(n - \frac{1}{2}\right)x}{\sin \frac{x}{2}} dx \quad \dots (2)$$

Equation (1) - (2) gives $I_{n+1} + I_{n-1} - 2I_n = 0 \Rightarrow I_{n-1}, I_n, I_{n+1}$ are in A.P.

Section – C

52. 00081.00

53. 00144.00

Sol. (for Q.52-53)

$$y_n = y_{n-1} + y_{n-2} + y_{n-3}$$

$$x_n = x_{n-1} + x_{n-2}$$

54. 05050.00

55. 02728.00

Sol. (for Q. 54-55)

$$\text{Let } A = C + 3I$$

$$\Rightarrow C = \begin{bmatrix} 0 & 0 & 0 \\ 6 & 0 & 0 \\ 9 & 6 & 0 \end{bmatrix}; C^2 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 36 & 0 & 0 \end{bmatrix}; C^3 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\left(\frac{A}{3}\right)^{50} = \left(I + \frac{C}{3}\right)^{50} = {}^{50}C_0 I + {}^{50}C_1 \left(\frac{C}{3}\right) + {}^{50}C_2 \left(\frac{C}{3}\right)^2$$

$$b_{31} = 5050$$

$$B + I = \left(\frac{A}{3}\right)^{50} = \begin{bmatrix} 1 & 0 & 0 \\ 100 & 1 & 0 \\ 5050 & 100 & 1 \end{bmatrix}$$

$$\text{Det}(A \cdot \text{Adj } 2(B + I)) = \text{Det } A \cdot (\text{Det } 2(B + I))^2 = 27.64 = 2728$$

56. 00084.00

57. 00007.27

Sol. (for Q. 56-57)

$$a + b - c = 6 \quad \dots (1)$$

$$\text{and } c^2 = (a + b)^2 - 2ab$$

$$\Rightarrow a + b + c = \frac{ab}{3} \quad \dots (2)$$

$$\Rightarrow 2(a + b) = \frac{ab}{3} + 6 \Rightarrow b = \frac{5(a - b)}{a - 6} = 6 + \frac{18}{a - 6} > a$$

$$\Rightarrow a \in (6, 6 + 3\sqrt{2}) ; a = 7, b = 24, a = 8, b = 15, a = 9, b = 12$$